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Intra- and Interspecific relationships between tadpoles and Insects in a temporary pond, southeastern Brazil.

Grupo Taxonômico: Anfíbios - Anuros **Categoria:** Ecologia/ História Natural

There are few studies about anuran communities in southeastern Brazil concerning the larval phase. Dos Santos et al. studied about the co-occurrence of tadpoles and aquatic insects in temporary ponds, and found that both abundance and distribution of predatory insects determined the microhabitat use by tadpoles. The aim of this study was to describe the local assemblage of tadpoles in temporary ponds in Fazenda Experimental de Lambari-MG (FELB-EPAMIG), identifying the potential predators and competitors, establishing their inter-relationships with abiotic factors and the patterns of microhabitat use by species.

Two temporary ponds were surveyed in the vicinity of FELB, one of them is a seepage pond in the road, and the other is a rice field. Voucher specimens were collected to species identification and after fixed and disposed in 5% formalin, being deposited in the herpetological collection of Laboratório e Museu de Zoologia da UNIFAL-MG. To establish the predation relationships in the surveyed habitats, we made essays as done by Heyer et al. With this data, allied to the microhabitat occupation we made a phenogram and a scheme of possible trophic relationships between tadpoles and insects. The relationships of competition were inferred after measurements of niche overlap.

The niche overlap data and the species pairs from the phenogram permit us to make inferences about the ecological likeness of species. From the cluster analysis (UPGMA), two great groups arose: one with *Scinax fuscovarius* and *S. fuscomarginatus* very close, with a close relationship to *Dendropsophus minutus* (reflecting the nektonic guild), and the other that contains *Leptodactylus fuscus* and *Physalaemus cuvieri*, followed by *D. rubicundulus* (grouping the benthic tadpoles). The niche overlap measurement delimited two species pairs with overlap indexes equal to 0. 816 (*P. cuvieri* - *D. rubicundulus* and *L. fuscus* - *P. cuvieri*). The data from the predation essays showed that the tadpoles of *L. fuscus* and *D. minutus* were fully palatable to the larvae of *Thermonectus* (Dytiscidae) and *Berosus* sp. (Hydrophilidae), and to adults of *Belostoma plebejum*. From the field observations, is expected that the tadpoles are a common prey of *Tramea* sp larvae (Odonata: Libellulidae).

The insects showed be an important component influencing the predation on tadpoles, hence limiting their distribution. Due the high index of niche overlap, *L. fuscus* - *P. cuvieri*, *D. rubicundulus* - *P. cuvieri* e *D. rubicundulus* - *L. fuscus* would be the species pairs most promising for future investigations on competitive interactions.

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